INCH-POUND

MIL-PRF-1/1339E
17 March 2008
SUPERSEDING
MIL-PRF-1/1339D
2 May 2003

PERFORMANCE SPECIFICATION SHEET

ELECTRON TUBE, CATHODE RAY

TYPE 5AMP1A

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the electron tube described herein shall consist of this document and MIL-PRF-1.

DESCRIPTION: Electrostatic deflection and focus.

See figure 1.

Mounting position: Any.

ABSOLUTE RATINGS:

Parameter: Unit:	Ef V dc	Ec1 V dc	ed v	Eb1 V dc	Eb2 V dc	Rg MegΩ	Zd MegΩ	Ehk V dc
Maximum: Minimum:	6.9 5.7	0 -200	1,320 ----	1,650 ----	6,600 ----	1.5 ----	1.0 ----	±180 ----
Test condition:	6.3	Adjust	----	Focus	2,500	----	----	----

GENERAL:

Qualification - Required.

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TABLE I. Testing and inspection.

Inspection	MIL-STD-1311 Method	Conditions	Acceptance Level 4/	Inspection level or code	Symbol	Limits		Unit
						Min	Max	
Qualification								
Vibration	5111		----	----	Width	----	1	mm
Pressure (implosion)	1141		----	----	----	----	----	----
Cathode illumination	5216		----	----	----	----	----	----
Direct-interelectrode capacitance	1331	g1 to all	----	----	Cg1	----	5.5	pF
		k to all	----	----	Ck	----	4.9	pF
		D1 to D2	----	----	C1D2	----	3.9	pF
		D3 to D4	----	----	C3D4	----	2.5	pF
		D1 to all	----	----	CD1	----	7.9	pF
		D2 to all	----	----	CD2	----	7.4	pF
		D3 to all	----	----	CD3	----	5.5	pF
		D4 to all	----	----	CD4	----	4.8	pF
Deflection-factor uniformity	5248		----	----	----	----	1	%
Neck and bulb alinement (electrostatic types)	5101		----	----	Dia	----	2.25	in.
Conformance inspection, part 1								
Voltage breakdown	5201		0.65	II	----	----	----	----
Voltage breakdown (electrostatic types)	5201		0.65	II	----	----	----	----
Gas “cross” (electrostatic deflection)	5206	Light = 15 fL ($51.4\ cd/m^2$) 1/	0.65	II	----	----	----	----
Screen and faceplate blemishes	5106		0.65	II	----	----	----	----
Light output	5221	Eb = 2500 V	0.65	II	----	15	----	fL
Modulation	5223	Light = 15 fL ($51.4\ cd/m^2$)	0.65	II	ΔEc	----	45	V dc
Spot position (electrostatic deflection)	5231		0.65	II	----	----	16	mm
Spot displacement (leakage)	5231		0.65	II	Displ	----	10	mm
Grid-cutoff voltage	5241		0.65	II	Eco	-34	-56	V dc
Grid No. 1 leakage current	5251		0.65	II	----	----	----	----
Anode No. 2 leakage current	5251		0.65	II	----	----	----	----
Pattern distortion	----	2/	0.65	II	----	----	----	----

See footnotes at end of table.

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TABLE I. Testing and inspection - Continued.

Inspection	MIL-STD-1311 Method	Conditions	Acceptance Level 4/	Inspection level or code	Symbol	Limits Min	Limits Max	Unit
Conformance inspection, part 1 - Continued								
Centering of scan	----	Focused trace; Ec1 at 10 V drive 3/ 1D2 3D4	 0.65 0.65	 II II	 ---- ----	 ±2.00 ±1.25	 ---- ----	 in. In.
Conformance inspection, part 2								
Heater current	1301		----	----	If	540	660	mA
Electrode current (anode No.1)	5201	Ec1 = 0	----	----	Ib1	-15	15	µA dc
Electrode current (anode No. 2)	5201	Light = 15 fL (51.4 cd/m^2)	----	----	Ib2	----	800	µA dc
Electrode current (cathode)	5201	Light = 15 fL (51.4 cd/m^2)	----	----	Ik	----	1,000	µA dc
Base alinement (electrostatic types)	5101	+1D2, pin No. 5	----	----	----	----	----	----
Angle between traces	5101		----	----	----	89.2	90.8	Degrees
Neck and base alinement (electrostatic types)	5101		----	----	----	----	----	----
Stray light emission (conventional types)	5216	Eb2 = 6,600 V	----	----	----	----	----	----
Line width “A” (electrostatic deflection)	5226	Light = 15 fL (51.4 cd/m^2)	----	----	Width	----	0.8	mm
Line width “B” (electrostatic deflection)	5226	Light = 15 fL (51.4 cd/m^2)	----	----	Width	----	0.85	mm
Focusing voltage at cutoff	5246		----	----	Eb1	----	300	V dc
Focusing voltage (zero-bias)	5246		----	----	Eb1	0	----	V dc
Deflection factor	5248	1D2	----	----	DF	40	50	V dc/in.
Deflection factor	5248	3D4	----	----	DF	20	25	V dc/in.
Heater-cathode leakage current	5251		----	----	----	----	----	----

See footnotes at end of table.

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TABLE I. Testing and inspection - Continued.

Inspection	MIL-STD-1311 Method	Conditions	Acceptance Level 4/	Inspection level or code	Symbol	Limits		Unit
						Min	Max	
Conformance inspection, part 3								
Life-test provisions	----	Group C Eb2 = 2,500 V Light = 15 fL (51.4 cd/m^2)	----	----	t	500		hrs.
Life-test end points	----	Same conditions as initial tests	----	----	Width	----	0.80	mm
		Line width "A"	----	----	Width	----	0.85	mm
		Line width "B"	----	----	ΔEc	----	45	V dc
		Modulation						

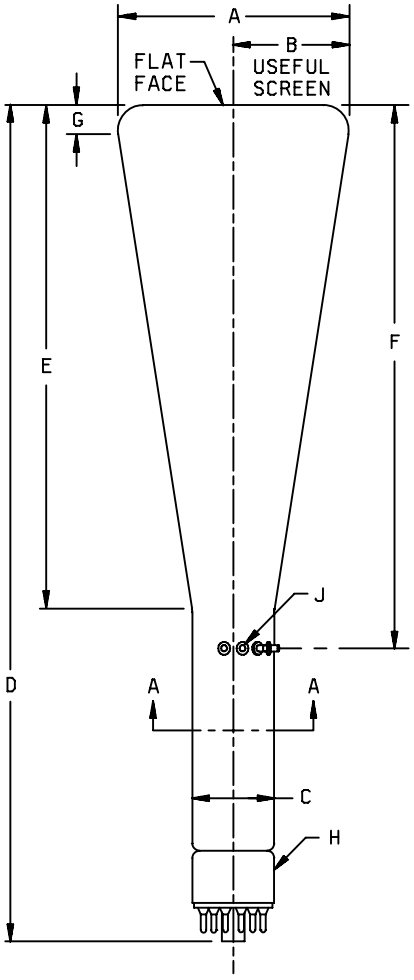
1/ This test to be performed at the conclusion of the holding period.

2/ All portions of a raster pattern, adjusted so its widest points just touch the sides of a 2.5 by 4.0-inch (63.5 by 101.6 mm) rectangle, will fall within the area bounded by the 2.5 by 4.0-inch (63.5 by 101.6 mm) rectangle and an inscribed 2.420 by 3.912-inch (61.47 by 99.36 mm) rectangle.

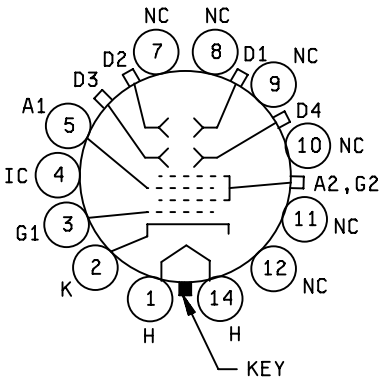
3/ The useful scan in the 1D2 and 3D4 directions is measured from the tube face center.

4/ This specification sheet uses accept on zero defect sampling in accordance with MIL-PRF-1, table III.

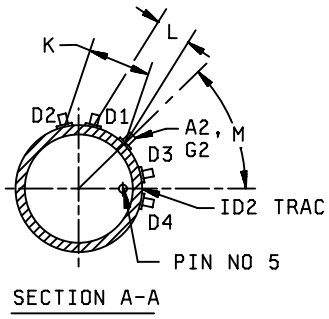
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Ltr	Dimensions			
	Millimeters		Inches	
Conformance inspection, part 2				
	Min	Max	Min	Max
A	130.96	135.74	5.156	5.344
B	57.15	----	2.250	----
C	49.23	52.37	1.938	2.062
D	441.33	454.03	17.375	17.875
E	----	228.60	----	9.000
F	315.26	332.28	12.412	13.082
K	26.92	29.97	1.060	1.180
L	12.70	14.22	0.500	0.560
M	40°	50°	40°	50°
Conformance inspection, part 3 (periodic check)				
H	Base: B12-37 (EIA)			
J	Caps: J1-25 (EIA)			
Reference dimensions (see note)				
G	22.23		0.875	



14 U
BOTTOM VIEW OF BASE



SECTION A-A

NOTE: Reference dimensions are for information only and are not required for inspection purposes.

FIGURE 1. Outline drawing of electron tube type 5AMP1A

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Referenced documents. In addition to MIL-PRF-1, this document references the following:
MIL-STD-1311

Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extent of the changes.

Custodians:
Army - CR
Navy - EC
DLA- CC

Preparing activity:
DLA - CC

(Project 5960-2008-024)

Review activities:
Army - MI
Navy - AS, CG, MC, OS, SA

NOTE: The activities listed above were interested in this document as of the date of this document. Since organizations and responsibilities can change, you should verify the currency of the information above using the ASSIST Online database at http://assist.daps.dla.mil.

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